CO226: Database Systems

Database Programming and Transactions

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1st July 2025

Database Programming

- Objective: to access a database from an application program (as opposed to interactive interfaces)
- Why? An interactive interface is convenient but not sufficient; a majority of database operations are made thru application programs (nowadays through web applications)

Database Programming Approaches

- Embedded commands: database commands are embedded in a general- purpose programming language
- Library of database functions: available to the host language for database calls; known as an API
- A brand new, full-fledged language (minimizes impedance mismatch) e.g. Oracle PL/SQL

Impedance Mismatch

- Incompatibilities between a host programming language and the database model, e.g.,
  - type mismatch and incompatibilities; requires a new binding for each language
  - set vs. record-at-a-time processing ..need special iterators to loop over query results and manipulate individual values

Steps in Database Programming

1. Client program opens a connection to the database server
2. Client program submits queries to and/or updates the database
3. When database access is no longer needed, client program terminates the connection

Database Programming in PHP (mysqli)

- Commonly used `mysqli` functions:
  - `mysqli_connect("hostname", "username", "password", "database")`: Connects to the MySQL server and selects the database.
  - `mysqli_query($connection, "query")`: Executes a query on the database connection.
  - `mysqli_fetch_assoc($result)`: Fetches one row from the result as an associative array.
  - `mysqli_error($connection)`: Returns a string description of the last error for the database connection.
  - `mysqli_close($connection)`: Closes the database connection.


```php
<?php
// connect to MySQL using mysqli
$connect = mysqli_connect("localhost", "root", "", "company");

// check connection
if (!$connect) {
    die("Connection failed: " . mysqli_connect_error());
}

// query the database
$query = "SELECT fname, lname FROM employee";
$results = mysqli_query($connect, $query);

if (!$results) {
    die("Query failed: " . mysqli_error($connect));
}

// display the results
while ($row = mysqli_fetch_assoc($results)) {
    echo $row['fname'] . " - " . $row['lname'] . "<br>";
}

// close the connection
mysqli_close($connect);
?>
```

Database Stored Procedures

- Persistent procedures/functions (modules) are stored locally and executed by the database server (as opposed to execution by clients)
- Advantages:
  - if the procedure is needed by many applications, it can be invoked by any of them (thus reduce duplications)
  - execution by the server reduces communication costs
  - When security is an important issue (e.g. Bank), use stored procedures and functions for all common operations. Applications and users have no access to the database tables directly
  - enhance the modeling power of views

Stored Procedure Constructs

- A stored procedure

```
CREATE PROCEDURE procedure-name (params)
local-declarations
procedure-body;
```

- A stored function

```
CREATE FUNCTION fun-name (params) RETRUNS return-type
local-declarations
function-body;
```

- Calling a procedure or function

```
CALL procedure-name/fun-name (arguments);
```


SP Example in MySQL

- Write a procedure to calculate the averaged employee salary for the company.

```
delimiter |
create procedure avgsalary()
begin
  select avg(salary) as avg_sal
  from employee;
end |
delimiter ;

call avgsalary();
```


SP Example in MySQL

- Write a function to calculate the averaged employee salary for a given department.

```
delimiter |
create function depavgsalary(no int) returns decimal(8,2)
begin
  declare avg decimal(8,2);
  select avg(salary)
  from employee
  where dno=no
  into avg;
  return avg;
end |
delimiter ;

select depavgsalary(5);
```


Introduction to Transaction Processing

Introduction to Transaction Processing

- Single-User System
  - At most one user at a time can use the system.
- Multiuser System
  - Many users can access the system concurrently.
- Concurrency
  - Interleaved processing: Concurrent execution of processes is interleaved in a single CPU
  - Parallel processing: Processes are concurrently executed in multiple CPUs.

Introduction to Transaction Processing (2)

- A Transaction:
  - Logical unit of database processing that includes one or more access operations (read -retrieval, write - insert or update, delete).
  - A transaction (set of operations) may be stand-alone specified in a high level language like SQL submitted interactively, or may be embedded within a program.
- Transaction boundaries:
  - Begin and End transaction.
  - An application program may contain several transactions separated by the Begin and End transaction boundaries.

Simple Model of a Database

- A database is a collection of named data items
- Granularity of data - a field, a record , or a whole disk block (Concepts are independent of granularity)
- Basic operations are read and write
- read_item(X): Reads a database item named X into a program variable. To simplify our notation, we assume that the program variable is also named X.
- write_item(X): Writes the value of program variable X into the database item named X.

Read and Write Operations

- Basic unit of data transfer from the disk to the computer main memory is one block. In general, a data item (what is read or written) will be the field of some record in the database, although it may be a larger unit such as a record or even a whole block.
- read_item(X) command includes the following steps:
  - Find the address of the disk block that contains item X.
  - Copy that disk block into a buffer in main memory (if that disk block is not already in some main memory buffer).
  - Copy item X from the buffer to the program variable named X.

Read and Write Operations (cont.)

- write_item(X) command includes the following steps:
  - Find the address of the disk block that contains item X.
  - Copy that disk block into a buffer in main memory (if that disk block is not already in some main memory buffer).
  - Copy item X from the program variable named X into its correct location in the buffer.
  - Store the updated block from the buffer back to disk (either immediately or at some later point in time).

Two sample transactions

(a) T_1	(b) T_2
read_item (X);	read_item (X);
X:=X-N;	X:=X+M;
write_item (X);	write_item (X);
read_item (Y);	
Y:=Y+N;	
write_item (Y);	

Figure: Two sample transactions: (a) Transaction T1 (b) Transaction T2

Why Concurrency Control is needed

- The Lost Update Problem:
  - This occurs when two transactions that access the same database items have their operations interleaved in a way that makes the value of some database item incorrect.
- The Temporary Update (or Dirty Read) Problem:
  - This occurs when one transaction updates a database item and then the transaction fails for some reason.
  - The updated item is accessed by another transaction before it is changed back to its original value.
- The Incorrect Summary Problem:
  - If one transaction is calculating an aggregate summary function on a number of records while other transactions are updating some of these records, the aggregate function may calculate some values before they are updated and others after they are updated.

The lost update problem

T_1	T_2
read_item(X);	
$X := X - N$;	
	read_item(X);
	$X := X + M$;
write_item(X);	
read_item(Y);	
	write_item(X);
$Y := Y + N$;	
write_item(Y);	

Time

Item X has an incorrect val
its update by T_1 is *lost* (ove

The temporary update problem

T_1	T_2
read_item(X);	
$X := X - N$;	
write_item(X);	
	read_item(X);
	$X := X + M$;
	write_item(X);
read_item(Y);	

Time

Transaction T_1 fails and must change the value of X back to its old value; meanwhile T_2 has read the *temporary* incorrect value of X.

The incorrect summary problem

T_1	T_3
	$sum := 0$;
	read_item(A);
	$sum := sum + A$;
	⋮
read_item(X);	
$X := X - N$;	
write_item(X);	
	read_item(X);
	$sum := sum + X$;
	read_item(Y);
	$sum := sum + Y$;
read_item(Y);	
$Y := Y + N$;	
write_item(Y);	

T_3 reads X after N is subtracted and reads Y before N is added; a wrong summary is the result (off by N).

Why recovery is needed

- A computer failure (system crash): A hardware or software error occurs in the computer system during transaction execution. If the hardware crashes, the contents of the computer's internal memory may be lost.
- A transaction or system error: Some operation in the transaction may cause it to fail, such as integer overflow or division by zero. Transaction failure may also occur because of erroneous parameter values or because of a logical programming error. In addition, the user may interrupt the transaction during its execution.

Why recovery is needed (cont.)

- Local errors or exception conditions detected by the transaction: Certain conditions necessitate cancellation of the transaction. For example, data for the transaction may not be found. A condition, such as insufficient account balance in a banking database, may cause a transaction, such as a fund withdrawal from that account, to be canceled. A programmed abort in the transaction causes it to fail.
- Concurrency control enforcement: The concurrency control method may decide to abort the transaction, to be restarted later, because it violates serializability or because several transactions are in a state of deadlock.

Why recovery is needed (cont.)

- Disk failure: Some disk blocks may lose their data because of a read or write malfunction or because of a disk read/write head crash. This may happen during a read or a write operation of the transaction.
- Physical problems and catastrophes: This refers to an endless list of problems that includes power or air-conditioning failure, fire, theft, sabotage, overwriting disks or tapes by mistake, and mounting of a wrong tape by the operator.

Transaction and System Concepts

- A transaction is an atomic unit of work that is either completed in its entirety or not done at all.
  - For recovery purposes, the system needs to keep track of when the transaction starts, terminates, and commits or aborts.
- Transaction states:
  - Active state
  - Partially committed state
  - Committed state
  - Failed state
  - Terminated State

Recovery manager keeps track of the following operations:

- begin_transaction: This marks the beginning of transaction execution.
- read or write: These specify read or write operations on the database items that are executed as part of a transaction.
- end_transaction: This specifies that read and write transaction operations have ended and marks the end limit of transaction execution. At this point it may be necessary to check whether the changes introduced by the transaction can be permanently applied to the database or whether the transaction has to be aborted because it violates concurrency control or for some other reason.

Recovery manager keeps track of the following operations: (cont.)

- commit_transaction: This signals a successful end of the transaction so that any changes (updates) executed by the transaction can be safely committed to the database and will not be undone.
- rollback (or abort): This signals that the transaction has ended unsuccessfully, so that any changes or effects that the transaction may have applied to the database must be undone.

Recovery techniques use the following operators:

- undo: Similar to rollback except that it applies to a single operation rather than to a whole transaction.
- redo: This specifies that certain transaction operations must be redone to ensure that all the operations of a committed transaction have been applied successfully to the database.

State transition diagram illustrating the states for transaction execution

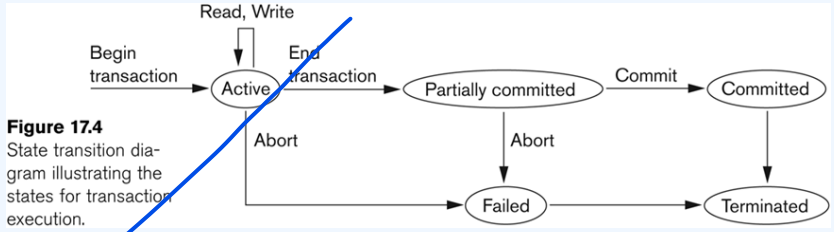


Figure 17.4
State transition diagram illustrating the states for transaction execution.

The System Log

- Log or Journal: The log keeps track of all transaction operations that affect the values of database items.
  - This information may be needed to permit recovery from transaction failures.
  - The log is kept on disk, so it is not affected by any type of failure except for disk or catastrophic failure.
  - In addition, the log is periodically backed up to archival storage (tape) to guard against such catastrophic failures.

The System Log (cont.)

- T in the following discussion refers to a unique transaction-id that is generated automatically by the system and is used to identify each transaction:
- Types of log record:
  - [start_transaction,T]: Records that transaction T has started execution.
  - [write_item,T,X,old_value,new_value]: Records that transaction T has changed the value of database item X from old_value to new_value.
  - [read_item,T,X]: Records that transaction T has read the value of database item X.
  - [commit,T]: Records that transaction T has completed successfully, and affirms that its effect can be committed (recorded permanently) to the database.
  - [abort,T]: Records that transaction T has been aborted.

Recovery using log records:

- If the system crashes, we can recover to a consistent database state by examining the log and using one of the Concurrency Control techniques discussed later.
  - Because the log contains a record of every write operation that changes the value of some database item, it is possible to undo the effect of these write operations of a transaction T by tracing backward through the log and resetting all items changed by a write operation of T to their old_values.
  - We can also redo the effect of the write operations of a transaction T by tracing forward through the log and setting all items changed by a write operation of T (that did not get done permanently) to their new_values.

Commit Point of a Transaction

- Definition a Commit Point:
  - A transaction T reaches its commit point when all its operations that access the database have been executed successfully and the effect of all the transaction operations on the database has been recorded in the log.
  - Beyond the commit point, the transaction is said to be committed, and its effect is assumed to be permanently recorded in the database.
  - The transaction then writes an entry [commit,T] into the log.
- Roll Back of transactions:
  - Needed for transactions that have a [start_transaction,T] entry into the log but no commit entry [commit,T] into the log.

Commit Point of a Transaction (cont.)

- Redoing transactions:
  - Transactions that have written their commit entry in the log must also have recorded all their write operations in the log; otherwise they would not be committed, so their effect on the database can be redone from the log entries.
- Force writing a log:
  - Before a transaction reaches its commit point, any portion of the log that has not been written to the disk yet must now be written to the disk.
  - This process is called force-writing the log file before committing a transaction.

ACID properties:

- Atomicity: A transaction is an atomic unit of processing; it is either performed in its entirety or not performed at all.
- Consistency preservation: A correct execution of the transaction must take the database from one consistent state to another.
- Isolation: A transaction should not make its updates visible to other transactions until it is committed; this property, when enforced strictly, solves the temporary update problem and makes cascading rollbacks of transactions unnecessary.
- Durability or permanency: Once a transaction changes the database and the changes are committed, these changes must never be lost because of subsequent failure.

Transaction schedule or history:

- When transactions are executing concurrently in an interleaved fashion, the order of execution of operations from the various transactions forms what is known as a transaction schedule (or history).
- A schedule (or history) S of n transactions $T_1, T_2, \ldots, T_n$:
  - It is an ordering of the operations of the transactions subject to the constraint that, for each transaction T_i that participates in S, the operations of T_1 in S must appear in the same order in which they occur in T_1.
  - Note, however, that operations from other transactions T_j can be interleaved with the operations of T_i in S.

Notation:

- For the purpose of recovery and concurrency control, read_item, write_item, commit and abort oprations are considered.
- The following notation is used.

  r for read_item
  w for write_item
  c for commit
  a for abort

Example: a

Time	T_1	T_2
	read_item(X); $X := X - N$;	
		read_item(X); $X := X + M$;
	write_item(X); read_item(Y);	
		write_item(X);
	$Y := Y + N$; write_item(Y);	

← Item X has an incorrect value because its update by T_1 is *lost* (overwritten).

- $T_1 : r_1(X); w_1(X); r_1(Y); w_1(Y);$
- $T_2 : r_2(X); w_2(X);$
- $S_a : r_1(X); r_2(X); w_1(X); r_1(Y); w_2(X); w_1(Y);$

Example: b

Time	T_1	T_2
	read_item(X); $X := X - N$; write_item(X);	
		read_item(X); $X := X + M$; write_item(X);
	read_item(Y);	

← Transaction T_1 fails and must change the value of X back to its old value; meanwhile T_2 has read the *temporary* incorrect value of X.

- $S_b : r_1(X); w_1(X); r_2(X); w_2(X); r_1(Y); a_1$

Conflicting operations

- Two operations in a schedule are said to be conflict if they satisfy all three of the following conditions:
  - they belong to different transactions
  - they access the same item X
  - at least one of the operations is a write_item(X)
- Example: consider schedule S_a
  - $r_1(X)\ w_2(X)$, $r_2(X)\ w_1(X)$, $w_1(X)\ w_2(X)$ conflict operations
  - $r_1(X)\ r_2(X)$, $w_2(X)\ w_1(Y)$ do not conflict

Schedules classified on recoverability:

- Recoverable schedule: Recovery is possible
  - One where no transaction needs to be rolled back.
  - A schedule S is recoverable if no transaction T in S commits until all transactions T' that have written an item that T reads have committed.
- Cascadeless schedule:
  - One where every transaction reads only the items that are written by committed transactions.

Schedules classified on recoverability:

- Schedules requiring cascaded rollback:
  - A schedule in which uncommitted transactions that read an item from a failed transaction must be rolled back.
- Strict Schedules:
  - A schedule in which a transaction can neither read or write an item X until the last transaction that wrote X has committed.

Examples

- $S_a^{'}$: $r_1(X); r_2(X); w_1(X); r_1(Y); w_2(X); c_2; w_1(Y); c_1$
  - $S_a^{'}$ is recoverable, even though it suffers from the lost update problem.
- S_c : $r_1(X); w_1(X); r_2(X); r_1(Y); w_2(X); c_2; a_1;$
  - S_c is not recoverable because T_2 reads item X from T_1, and then T_2 commits before T_1 commits.

Examples (cont.)

- $S_d : r_1(X); w_1(X); r_2(X); r_1(Y); w_2(X); w_1(Y); c_1; c_2;$
  - For the S_c schedule to be recoverable, the c_2 operation must be postponed until after T_1 commits as shown in S_d.
- $S_e : r_1(X); w_1(X); r_2(X); r_1(Y); w_2(X); w_1(Y); a_1; a_2;$
  - If T_1 aborts instead of commiting, then T_2 should also abort as shown in S_e.

Characterizing Schedules based on Serializability

- Serial schedule:
  - A schedule S is serial if, for every transaction T participating in the schedule, all the operations of T are executed consecutively in the schedule.
  - Otherwise, the schedule is called nonserial schedule.
- Serializable schedule:
  - A schedule S is serializable if it is equivalent to some serial schedule of the same n transactions.

Characterizing Schedules based on Serializability

- Result equivalent:
  - Two schedules are called result equivalent if they produce the same final state of the database.
- Conflict equivalent:
  - Two schedules are said to be conflict equivalent if the order of any two conflicting operations is the same in both schedules.
- Conflict serializable:
  - A schedule S is said to be conflict serializable if it is conflict equivalent to some serial schedule S'.

Characterizing Schedules based on Serializability

- Being serializable is not the same as being serial
- Being serializable implies that the schedule is a correct schedule.
- It will leave the database in a consistent state.
- The interleaving is appropriate and will result in a state as if the transactions were serially executed, yet will achieve efficiency due to concurrent execution.

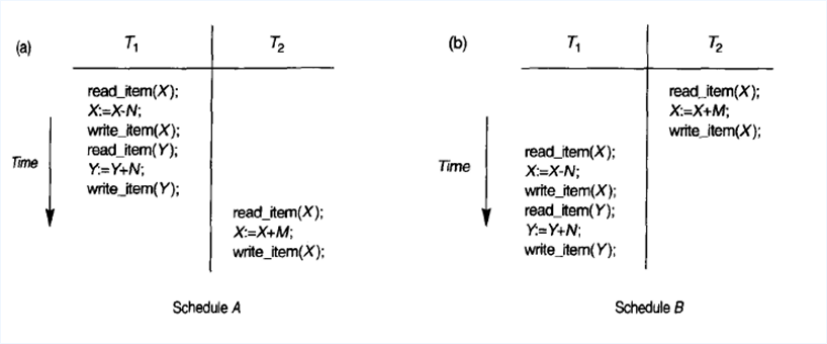
Characterizing Schedules based on Serializability

- Serializability is hard to check.
  - Interleaving of operations occurs in an operating system through some scheduler
  - Difficult to determine beforehand how the operations in a schedule will be interleaved.

Testing for conflict serializability:

- Algorithm:
  - Looks at only read_Item (X) and write_Item (X) operations
  - Constructs a precedence graph (serialization graph) - a graph with directed edges
  - An edge is created from T_i to T_j if one of the operations in T_i appears before a conflicting operation in T_j
  - The schedule is serializable if and only if the precedence graph has no cycles.

Constructing the Precedence Graphs



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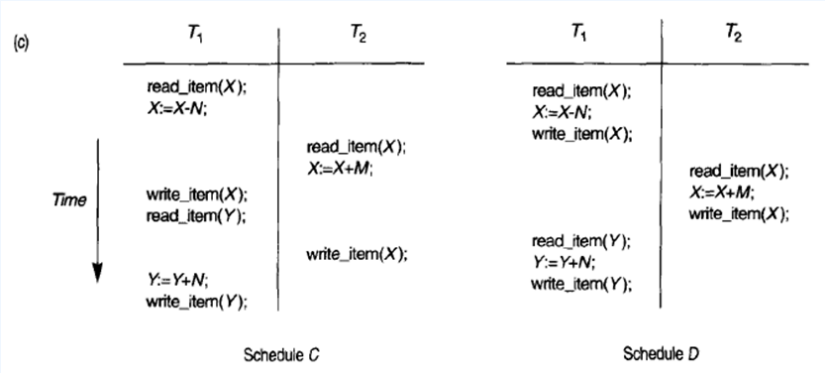

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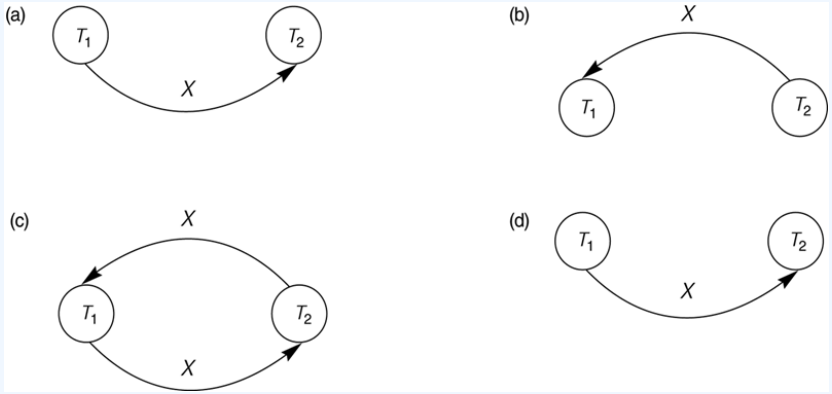

How Serializability is Used for Concurrency Control

- Being serializable is different from being serial
- Serializable schedule gives benefit of concurrent execution
  - Without giving up any correctness
- Difficult to test for serializability in practice
  - Factors such as system load, time of transaction submission, and process priority affect ordering of operations
- DBMS enforces protocols
  - Set of rules to ensure serializability

MySQL Transaction Support

- Transactions begin with:
  - `START TRANSACTION;` or `BEGIN;`
- Alternatively, disable automatic commits:
  - `SET autocommit = 0;`
  - Requires explicit `COMMIT;` or `ROLLBACK;` for every transaction
- Every transaction must end explicitly with:
  - `COMMIT;` to save changes
  - `ROLLBACK;` to undo changes
- Access mode can be specified as:
  - `START TRANSACTION READ ONLY;`
  - `START TRANSACTION READ WRITE;`

MySQL Transaction Support (cont.)

- Storage Engine Support:
  - InnoDB – Fully supports transactions, foreign keys, and row-level locking
  - MyISAM – Does not support transactions or foreign keys
  - Use `SHOW TABLE STATUS;` to check storage engine
- Best Practices:
  - Use InnoDB for reliable transactional behavior
  - Always close transactions with `COMMIT` or `ROLLBACK`

Indexing in MySQL

- A data structure that improves the speed of data retrieval
- Types of Indexes:
  - `PRIMARY KEY` – Automatically creates a unique index
  - `UNIQUE` – Ensures all values in the column are distinct
  - `INDEX` – Speeds up queries (e.g., on `WHERE` clauses)
  - `FULLTEXT` – Used for text search
  - `SPATIAL` – For geographic data (in MyISAM or InnoDB)
- Creating an Index:
  - `CREATE INDEX idx_name ON table_name(column);`
- Considerations:
  - Speeds up `SELECT`, slows down `INSERT/UPDATE/DELETE`
  - Use indexes on columns used in `WHERE`, `JOIN`, or `ORDER BY`

Query Optimization and EXPLAIN in MySQL

- Query Optimization:
  - Process of improving SQL query performance
  - Use proper indexes to reduce full table scans
  - Avoid using `SELECT *`; fetch only required columns
  - Use conditions on indexed columns in `WHERE` clauses
  - Minimize subqueries and use `JOIN`s wisely
- Using `EXPLAIN`:
  - Shows how MySQL executes a query
  - Helps identify bottlenecks and missing indexes
  - Syntax: `EXPLAIN SELECT ...;`
  - Key fields in output:
    - `type` – join type (e.g., ALL, index, ref)
    - `key` – index used
    - `rows` – estimated number of rows scanned
    - `Extra` – additional info (e.g., Using index)